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A self-consistent three-dimensional model of the arc, electrode and weld pool in gas–metal arc welding

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Abstract

The development of a three-dimensional computational model of gas–metal arc welding is described. The wire electrode, arc plasma and weld pool are included in the computational domain self-consistently. The model takes into account the motion of the electrode, flow in the weld pool, deformation of the weld-pool surface and the influence of metal droplet transfer. Results are presented for welding of an aluminium alloy. The current density distribution at the interface between the arc and the weld pool is strongly dependent on the surface profile of the weld pool. This in turn affects the temperature distribution in the weld pool. The momentum transferred by the droplet affects the direction of flow in the weld pool, and together with the energy transfer, increases the weld-pool depth. The results demonstrate the importance of including the arc plasma in the computational domain. Fair agreement is found between a measured weld profile and the predictions of the model. Inclusion of the influence of metal vapour in the model is expected to improve the agreement.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Arc welding is a very widely used process for joining metals. Gas–metal arc welding (GMAW; also known as metal–inert-gas or metal–active-gas welding) is the most common industrial arc-welding technique. An electric arc is struck between a continuously fed metal wire electrode and the workpiece. The usual polarity is electrode positive; i.e. the wire electrode is the anode and the workpiece is the cathode. Droplets of molten metal detach and fall to the workpiece, which is partially melted by the arc to form the weld pool.

GMAW is well suited to automation, and in many industries welding robots are integrated into the production line. Modern techniques such as pulse-spray transfer, in which the arc current is pulsed and one droplet detached during each pulse, allow good control over the heat transfer to the workpiece. Nevertheless, there are some situations, particularly when metals with high conductivity such as aluminium are being welded, in which reproducibility and reliability problems occur. The development of computational models of arc welding that are capable of predicting the depth and shape of the weld pool will improve understanding of such problems, and allow the optimum welding parameters for production of good-quality welds to be determined.

There are several difficulties inherent in modelling of GMAW. The motion of the wire electrode and arc relative to the workpiece has to be taken into account. The geometry is three-dimensional, even for bead-on-plate (i.e. flat workpiece) geometry, due to the motion of the arc. Inclusion of the plasma in the computational domain is essential to allow accurate determination of the heat flow, arc pressure, current density and shear stress at the weld-pool surface, which in turn are critical in determining the geometry of the weld pool [1]. The flow in the molten weld pool requires careful treatment [2]. The surface of the weld pool is not flat, in particular because of the addition of molten metal by the droplets. The momentum and energy transferred to the weld pool by the droplets also have to be taken into account.

Computational models of arc welding of varying degrees of sophistication have previously been developed. Many models of weld-pool phenomena (e.g. [3–6]) have not included the wire electrode and arc plasma in the computational domain,

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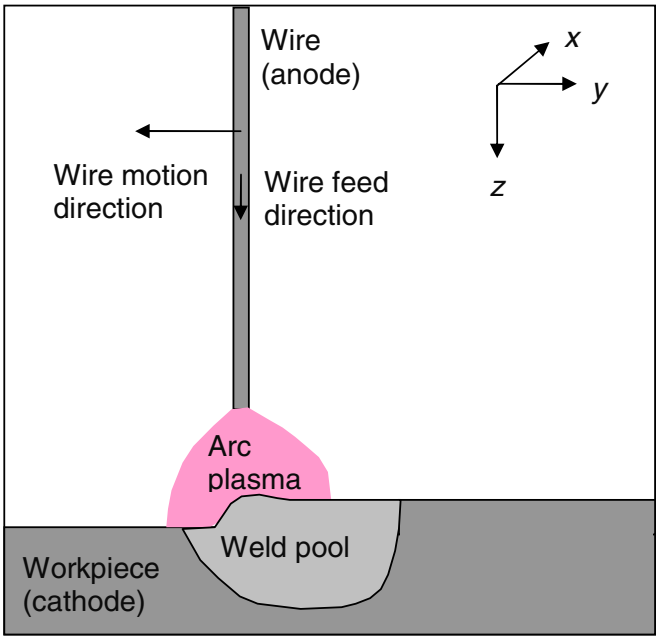


Figure 1. Schematic of the computational domain. The boundaries between the arc plasma and the wire electrode, the arc plasma and the workpiece, and the weld pool and the solid region of the workpiece are shown.

but instead have relied on estimated parameters [3–5] or measured parameters [6] at the weld-pool surface. Others have considered both the arc plasma and weld pool, but not self-consistently (e.g. [7]), or assumed that the surface of the weld pool is flat (e.g. [2]). The formation of molten metal droplets at the wire electrode in GMAW [8] and their transfer into the weld pool [9–11] have been modelled; these studies have used two-dimensional axisymmetric geometry, and have therefore not considered the motion of the arc along the weld seam. A number of three-dimensional models of arcs have been developed (e.g. [12, 13]) that have not included the weld pool. Recently, Xu *et al* [14] published details of a three-dimensional model that included the arc, the formation of droplets and flow in the weld pool. However, the coupling between the arc and weld pool was not fully self-consistent, and the heat transfer in the wire electrode was not calculated. Further, the focus of the paper was the transit of a single droplet through the arc, so the weld-pool geometry was not discussed.

In this paper, a comprehensive treatment of GMAW that addresses all the difficulties listed above is described, and representative results are presented. In particular, the influence of the surface deformation of the weld pool, and the momentum and energy transferred by the droplets, is examined. The importance of including the arc plasma in the computational domain is demonstrated.

2. Methods

2.1. Governing equations

The computational domain is shown schematically in figure 1. The domain includes the wire electrode, the arc plasma and workpiece. Four different phases of matter (solid, liquid, gas and plasma) are present in these regions; nevertheless, the same set of equations can be used for the whole domain. The equations are those of computational fluid dynamics, modified to take into account the high temperatures that occur in plasmas, and the flow of electric current through the electrode, plasma and workpiece. This requires the solution of an energy conservation and current continuity equation in addition to the usual Navier–Stokes (mass continuity and momentum conservation) equations. In addition, a 'magnetic pinch' or $\boldsymbol{j} \times \boldsymbol{B}$ term is added to the momentum conservation equation to take into account the pressure gradient produced by the self-magnetic field that is induced by the current. To take into account the motion of the wire electrode and arc relative to the workpiece, the equations are written in the frame of reference of the wire electrode, moving at velocity $\boldsymbol{V}$ with respect to the workpiece [15]. Only steady-state calculations are reported here, so the equations are given in steady-state form. This restriction means that the model is limited to the treatment of welding with constant speed, and constant average current. The influence of droplet formation and transfer is treated on a time-averaged basis, as discussed in sections 2.3 and 4.

The mass continuity equation is

$$\nabla \cdot (\rho \boldsymbol{v}) = 0. \tag{1}$$

The momentum conservation equation is

$$\nabla \cdot [\rho(\boldsymbol{v} - \boldsymbol{V})(\boldsymbol{v} - \boldsymbol{V})] = -\nabla P + \nabla \cdot \boldsymbol{\tau} + \boldsymbol{j} \times \boldsymbol{B} + \rho g. \tag{2}$$

The energy conservation equation is

$$\nabla \cdot [\rho(\boldsymbol{v} - \boldsymbol{V})h] = \frac{j^2}{\sigma} + \nabla \cdot \left(\frac{k}{c_p} \nabla h\right) - \frac{5k_\mathrm{B}}{2ec_p} \boldsymbol{j} \cdot \nabla h - U. \tag{3}$$

The current continuity equation is

$$\nabla \cdot (\sigma \nabla \phi) = 0. \tag{4}$$

In the above equations, $\boldsymbol{v}$ is the fluid flow velocity, h is the enthalpy, P is the pressure, g is the acceleration due to gravity, k_B is Boltzmann's constant, e is the electronic charge and μ_0 is the permeability of free space. The viscous stress tensor $\boldsymbol{\tau}$ is given by

$$\begin{aligned} \tau_{ii} &= \eta \left(2\frac{\partial v_i}{\partial x_i} - \frac{2}{3}\nabla \cdot \boldsymbol{v}\right), \\ \tau_{ij} &= \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right), \qquad i \neq j, \end{aligned} \tag{5}$$

where v_i and x_i are, respectively, the components of the velocity and position in either the $i = x$, y or z direction.

The thermophysical properties are the mass density ρ, the viscosity η, the specific heat at constant pressure c_p, the thermal conductivity k, the electrical conductivity σ and the net radiative emission coefficient U. The enthalpy h is the integral of specific heat with respect to temperature. The current density $\boldsymbol{j}$ is related to the electric potential ϕ by

$$\boldsymbol{j} = -\sigma \nabla \phi. \tag{6}$$

The magnetic field strength $\boldsymbol{B}$ is calculated using $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, with the magnetic vector potential $\boldsymbol{A}$ determined by solving

$$\nabla^2 \boldsymbol{A} = -\mu_0 \boldsymbol{j}. \tag{7}$$

The different phases are differentiated by their different thermophysical properties. In the solid regions, the mass continuity and momentum conservation equations are not required since there is no flow.

2.2. Boundary conditions

Boundary conditions have to be imposed in order to solve the equations. The conditions applied at the external boundaries of the computational domain are given in section 2.2.1. The internal boundaries are then discussed. The heat and momentum transfer at the interface between the arc and the electrode and workpiece is considered in section 2.2.2, and the heat transfer at the interface between the molten weld pool and the solid region of the workpiece in section 2.2.3. The treatment of the free surface between the arc plasma and the weld pool is considered in section 2.2.4.

2.2.1. External boundaries. The temperature was set to 300 K at all external boundaries, except the bottom of the workpiece. Here the heat flux coefficient to the surroundings was set to 50 W m^{-2} K^{-1}, consistent with unforced air cooling. The velocity was set to zero at external boundaries, except for a small region at the top of the electrode to simulate gas inflow through a nozzle. The electric potential was set to zero at the bottom of the workpiece. Its gradient was set to zero at other external boundaries, except at the top of the electrode, where $\partial\phi/\partial z = I/(\pi\sigma_e r_e^2)$, I being the arc current, σ_e the electrical conductivity of the electrode and r_e the electrode radius, to ensure the correct arc current flows in the electrode. The gradient of the magnetic potential was set to zero at all external boundaries. The pressure need only be defined at one point, and was set to 1 atm at one of the top corners of the computational domain. No axis of symmetry is assumed.

2.2.2. Heat and momentum transfer between arc plasma and the electrode and workpiece. Internal boundaries between the electrode and plasma, and workpiece and plasma, require special treatment. The workpiece is a non-thermionic cathode, and the mechanism by which electrons are transferred from such cathodes to the arc is not well understood. The model of Lowke and Tanaka [16], which postulates that electrons are emitted as a consequence of the energy transferred to the cathode by excited atoms from the arc, is used. In this case, an additional term is added to the energy conservation equation (3) at computational cells at the edge of workpiece:

$$S_w = j_w(V_w - \varphi_{ww}) - \varepsilon_w a T_w^4. \tag{8}$$

The first term represents the heating effect of the excited atoms that cause electrons to be emitted from the workpiece and accelerated by the sheath voltage, less the energy required to emit the electrons. In (8), j_w is the local current density, V_w is the cathode fall voltage and φ_{ww} is the work function of the workpiece material. Total electrode fall voltages between 16.5 and 16.7 V have been measured for iron, copper and titanium electrodes [17]; assuming the anode fall voltage is about 1.5 V, and that the value for aluminium will be similar, $V_w = 15$ V is obtained. The second term, which is generally negligible, is the blackbody radiation loss, with ε_w the emissivity of the workpiece material, a the Stefan–Boltzmann constant and T_w the local workpiece temperature. In addition, thermal conduction occurs from the arc to the workpiece; this is included in (3), and is usually less important than the electron heating.

The boundary between the plasma and the wire electrode is treated using the 'LTE diffusion model' of Lowke and Tanaka [18]. For computational cells lying on this boundary, an additional term is added to the energy conservation equation (3):

$$S_e = j\varphi_{we} - \varepsilon_e a T_e^4, \tag{9}$$

where the first term represents the heating effect of electrons absorbed at the electrode, with φ_{we} the work function of the electrode material, and the second term represents radiative cooling of the electrode; ε_e is the emissivity of the electrode material and T_e is the surface temperature of the electrode.

Two boundary conditions are required to treat momentum transfer across the boundary between the weld pool and the plasma. These are, respectively,

$$\tau_a - \tau_w + \frac{\partial\gamma}{\partial T}\nabla_S T = 0 \tag{10}$$

and

$$P_a - P_w + \gamma\kappa = 0 \tag{11}$$

parallel and perpendicular to the boundary. The first two terms in (10) are, respectively, the shear stresses on the arc and weld-pool sides of the boundary, and the third term is the Marangoni term, which describes the variation of the surface tension γ due to the gradient of temperature in the surface layer of the weld pool. The operator ∇_S denotes the gradient parallel to the surface. The factor $\mathrm{d}\gamma/\mathrm{d}T$ is negative for most pure metals, but can be positive in the presence of impurities such as oxygen or sulfur. The first two terms in (11) are, respectively, the pressures at the plasma and weld-pool sides of the boundary; the pressure difference and the surface tension determine the curvature κ of the boundary.

2.2.3. Liquid–solid interface. The boundary between the solid and liquid regions of the workpiece is, for a pure metal, defined by the melting temperature of the metal. For an alloy, the liquidus and solidus temperatures are different; the region at intermediate temperatures is termed the mushy zone. In both cases, the latent heat of melting of the metal has to be treated carefully, since there is a large change in enthalpy in a small region. The current results are calculated neglecting the influence of the mushy zone; i.e. assuming the solidus temperature is equal to the liquidus temperature. The method proposed by Voller and Prakash [21] was used to treat the latent heat of melting in a manner that ensures numerical stability at the liquid–solid boundary.

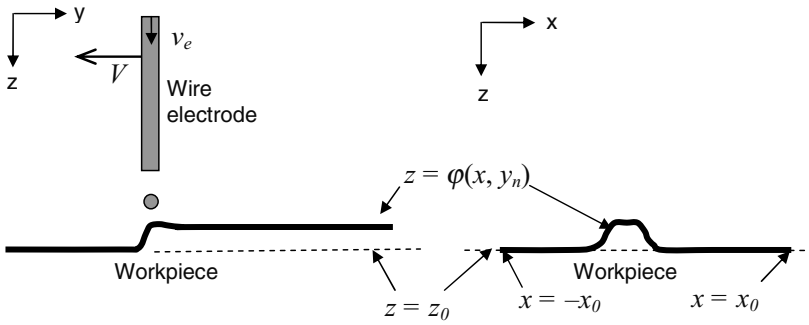


Figure 2. Cross-section at $x = 0$ in the y–z plane (left) and at $y = y_n$ in the x–z plane (right), showing the reinforcement of the workpiece by the droplets.

2.2.4. Free surface between weld pool and arc plasma. Treatment of the non-horizontal free surface between the weld pool and the plasma is a difficult problem requiring advanced techniques. The most widely used approach is the volume-of-fluid method [24]. However, this method requires very small time steps, since the free surface may not cross more than one control volume during each step. Instead, an equilibrium surface method, derived by Kim and Na [25] and by Wu *et al* [26], is used to treat the deformation of the surface of the weld pool due to the arc and droplet pressure and the addition of mass through droplet transfer. In this method, the weld-pool surface profile is determined by assuming it has reached equilibrium. The surface profile is calculated by minimizing the total surface energy, taking into account the surface tension energy, the gravitational potential energy and the work performed on the surface by the arc pressure. The equation for the free surface profile is

$$\begin{aligned}\gamma\{&[[1+(\partial\varphi/\partial y)^2](\partial^2\varphi/\partial x^2)\\&-2(\partial\varphi/\partial x)(\partial\varphi/\partial y)(\partial^2\varphi/\partial x\partial y)\\&+[1+(\partial\varphi/\partial x)^2](\partial^2\varphi/\partial y^2)]\\&\times[1+(\partial\varphi/\partial x)^2+(\partial\varphi/\partial y)^2]^{-\frac{3}{2}}\}\\&=\rho_{\mathrm{l}}g\varphi+P_{\mathrm{a}}+P_{\mathrm{d}}+\lambda,\end{aligned}\tag{12}$$

where $\varphi(x, y)$ is the surface height, $\gamma(x, y)$ is the surface tension, $\rho_{\mathrm{l}}(x, y)$ is the density of the weld-pool liquid, $P_{\mathrm{a}}(x, y)$ and $P_{\mathrm{d}}(x, y)$ are, respectively, the pressure applied by the arc and the droplets and λ is a constant (the Lagrange multiplier).

The Lagrange multiplier is determined by a constraint based on the volume of metal fed from the wire electrode. For a moving arc, this constraint is easily obtained from the wire electrode feed speed v_{e}, the radius of the electrode r_{e} and the welding speed V by integrating the height of the workpiece relative to the height before melting z_0 at a position y_n that is far enough behind the weld pool such that the weld pool has solidified. If the welding velocity is in the y-direction, then the condition is

$$V\int_{-x_0}^{x_0}[z_0-\varphi(x, y_n)]\,\mathrm{d}x=\pi r_{\mathrm{e}}^2 v_{\mathrm{e}}.\tag{13}$$

The integral on the left-hand side is just the cross-sectional area of the reinforcement; the integral is from one edge of the workpiece to the other. This is shown schematically in figure 2.

The surface curvature term in boundary condition (11) can cause numerical problems, in particular since it is difficult to calculate the curvature κ accurately. These can be minimized using the continuum surface force model developed by Brackbill *et al* [27]. Instead of applying the surface pressure boundary condition (11) at the discontinuity defined by the surface, a volume force term is added to the momentum conservation equations for computational cells lying within a transition region of finite thickness around the surface. This eliminates the need for accurate interface reconstruction. A further refinement adopted in the calculation of the curvature is the use of mollification, following the approach of Rudman [28].

The influence of aluminium vapour has not yet been included in the model; the importance of this assumption is discussed in section 4.

2.3. Treatment of droplets

The temperature and velocity of the droplets are tracked from their initial detachment to the bottom of the weld pool. The diameter of the droplets depends on the electrode diameter, feed velocity and droplet detachment frequency. The initial temperature is equated to the average temperature of the region at the tip of the wire electrode with volume equal to that of a droplet, and the initial velocity was determined using the formula given by Lin *et al* [22]. The increase in droplet temperature and velocity as the droplet falls through the arc is calculated using the method of Crowe *et al* [23], neglecting the influence of vaporization. The transfer of enthalpy and momentum from the plasma to the droplets, and subsequently from the droplets to the liquid metal in the weld pool, is calculated by averaging over time the change in droplet temperature and velocity. This is taken into account in the model by including additional source terms in the energy and momentum conservation equations in the control volumes through which the droplets travel.

2.4. Computational approach

Equations (1) to (4) and (7) were solved using the finite-volume method described by Patankar [19]. The equations were discretized on a three-dimensional non-uniform Cartesian grid. The results presented in this paper were obtained using

Table 1. Material properties of workpiece and wire.

		Solid		Liquid	
Property	Temperature	300 K	⩾1000 K	300 K	⩾1000 K
k (W m^{-1} K^{-1})		134	124	104	84
σ (S m^{-1})		3.6×10^7	8.2×10^6	3.6×10^7	8.2×10^6
η (kg m^{-1} s^{-1})		—	—	1.3×10^{-3}	1.3×10^{-3}
c_p (J kg^{-1} K^{-1})		900	900	1200	1200
ρ (kg m^{-3})		2700	2700	2700	2700
γ (N m^{-1})		—	—	0.620	
dγ/dT (N m^{-1} K^{-1})		—	—	-2.25×10^{-4}	

61 × 71 × 85 grid points in the x, y and z dimensions, respectively. The computational domain was approximately 40 mm × 50 mm × 55 mm with the grid size ranging from 0.03 mm near the boundaries between the plasma and the electrode and workpiece, as recommended by Lowke and Tanaka [18], to 5 mm at the edges of the domain. To avoid numerical difficulties, the momentum equations were solved on a staggered grid (at computational cell edges rather than centres). The mass continuity equation (1) was solved using the SIMPLEC algorithm [20], which allows calculation of a pressure field for which the velocities satisfy equation (1). Each of the eight other coupled partial differential equations (three each for momentum conservation and magnetic potential calculation, as well as energy conservation and current continuity) was solved by a combination of the tridiagonal matrix algorithm and the Gauss–Seidel point-by-point method [19].

2.5. Material properties and welding parameters

Calculations were performed for argon shielding gas and aluminium alloy AA 5754 wire electrode and workpiece. The thermodynamic and transport properties of argon are taken from the work of Murphy and Arundell [29], and the net radiative emission coefficients from the work of Cram [30]. The thermal conductivity, surface tension and its dependence on temperature, and liquidus temperature (875 K) for aluminium alloy AA 5754 were taken from [31]. Data for pure aluminium were used for the other properties (density, electrical conductivity, specific heat and viscosity) [32]. The metal property data are given in table 1; linear interpolation is used for temperatures between 300 and 1000 K, except for the surface tension, for which the value given is used at 875 K, and above this temperature $\gamma = \gamma(875\,\mathrm{K}) + \mathrm{d}\gamma/\mathrm{d}T\,(T - 875\,\mathrm{K})$ is used. The Boussinesq approximation is used in the weld pool, so it is assumed that the density of liquid metal is independent of temperature.

Calculations were performed with an initially flat workpiece (i.e. bead-on-plate welding), and for constant welding speed and arc current. The welding parameters considered are typical of one-drop-per-pulse welding of aluminium; these are listed in table 2. In one-drop-per-pulse welding, the arc current has a step waveform, with a background current and a pulse current. A droplet is detached from the wire during each pulse. Consistent with the time-averaging of the influence of the droplets on the arc and weld pool, a constant arc current (equal to the time-average of the step waveform) is used in the model. A flat wire tip is used.

Table 2. Parameters used for the calculations.

Parameter	Value
Arc current I	95 A
Wire electrode radius r_e	0.6 mm
Welding velocity V	15 mm s^{-1} in negative y-direction
Wire electrode feed rate v_e	72 mm s^{-1}
Droplet frequency	93 Hz
Workpiece thickness	3 mm

To allow comparison with measurements, in which the workpiece was clamped onto a metal bench, it is assumed in the model that the workpiece is attached to an aluminium slab of thickness 50 mm, with no thermal resistance at the interface.

3. Results

3.1. Weld-pool surface deformation and its influence

The transfer of metal from the wire electrode to the weld pool in the form of droplets leads to an increase in the height of the workpiece. As the wire moves horizontally with respect to the workpiece, the region of the workpiece heated by the arc also moves, and the previously molten region solidifies. This leaves a reinforced region, as was shown schematically in figure 2.

Figure 3 shows the calculated height of the workpiece surface. The workpiece is flat in the region in front of the arc. The influence of the droplets leads to an increase in surface height directly under the electrode. The boundary of the molten region, in which the workpiece height is free to change, is shown. Outside this region, the metal is frozen; the height of the reinforcement region is therefore fixed for $y \gtrsim 14$ mm. As can be seen from (13), the cross-sectional area of the reinforcement region is determined by the wire electrode radius, wire electrode feed rate and welding speed. In automated welding, these parameters are selected prior to welding being carried out. Nevertheless, the height profile of the reinforcement region does depend on the properties of the arc, droplets and weld pool.

For the parameters investigated here, the surface height is always positive. For larger arc currents, the flow velocity in the arc increases due to the larger magnetic pinch force, reaching about 250 m s^{-1} for a current of 200 A (compared with a maximum of 60 m s^{-1} for the 95 A used here). This

increases the arc pressure, and can lead to a depression in the region under the wire. For example, for an arc current of 200 A, a depression of maximum depth 0.3 mm and diameter around 3 mm is formed. However, the depression is filled by the flow in the weld pool beyond this region, and the reinforcement region becomes convex well before the weld pool solidifies.

A minor asymmetry in the surface height profile, with the peak height offset by just under 1 mm from the x-axis, is apparent in figure 3. As noted in section 2, symmetry is not enforced in the calculations. The asymmetry may be a consequence of numerical inaccuracies; in any case it does not have a significant influence on the weld-pool properties.

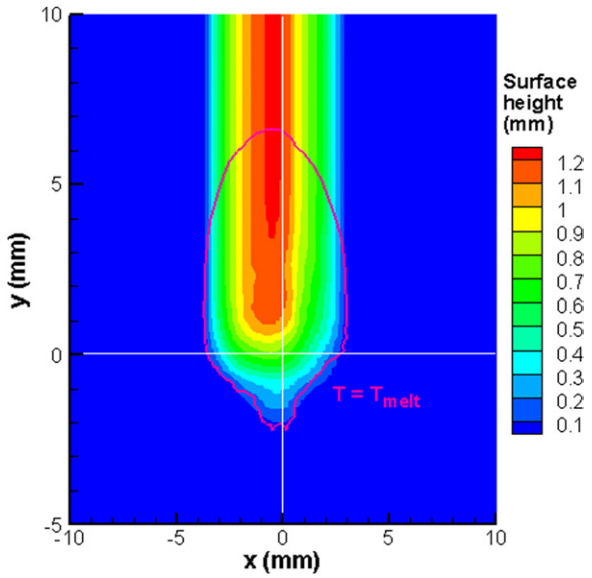


Figure 3. The height of the workpiece surface relative to its original height. The welding velocity is in the negative y-direction. The centre of the wire electrode is at $x = 0$, $y = 0$, at the intersection of the white lines. The region inside which the surface is above the melting temperature is shown by the closed curve labelled $T = T_{\text{melt}}$.

Figure 4 shows temperature distributions in two perpendicular cross-sections through the electrode, arc and workpiece. The arc temperature reached 17 500 K below the wire. The reinforcement of the weld pool due to the volume added by the droplets is clearly apparent. The hottest region in the weld pool is near the top of the reinforced region. The weld pool (delineated by the 875 K contour) is reasonably symmetric in the x–z plane, as expected. In the y–z plane, the weld pool lags the wire and arc; this is due to the motion of the wire in the negative y-direction. The greater the welding speed, the larger this effect becomes.

The current density distributions in a cross-section through the electrode, arc and workpiece are shown in figure 5. For comparison, results are also given for an artificial case, in which the weld-pool surface is forced to remain flat (that is, the volume added by the droplets is neglected). It is clear that the current density distribution is strongly affected by the increased height of the workpiece, and is consequently markedly asymmetric.

Computational models that do not include the arc self-consistently use estimated current density and heat flux distributions as boundary conditions at the workpiece surface. A typical approach [1] is to use a Gaussian distribution of current density and heat flux. For example, Zhang *et al* [33] used boundary conditions for the heat flux Q_{a}, current density j_{a} and arc pressure P_{a} at the weld-pool surface that assumed Gaussian distributions of these quantities:

$$Q_{\text{a}} = \frac{\alpha I V_{\text{a}}}{2\pi R_q^2} \exp\left(-\frac{r^2}{2R_q^2}\right) - h_{\text{c}}(T_{\text{w}} - T_{\text{a}}),$$

$$P_{\text{a}} = \frac{F}{2\pi R_{\text{p}}^2} \exp\left(-\frac{r^2}{2R_{\text{p}}^2}\right), \tag{14}$$

$$j_{\text{a}} = \frac{I}{2\pi R_q^2} \exp\left(-\frac{r^2}{2R_q^2}\right),$$

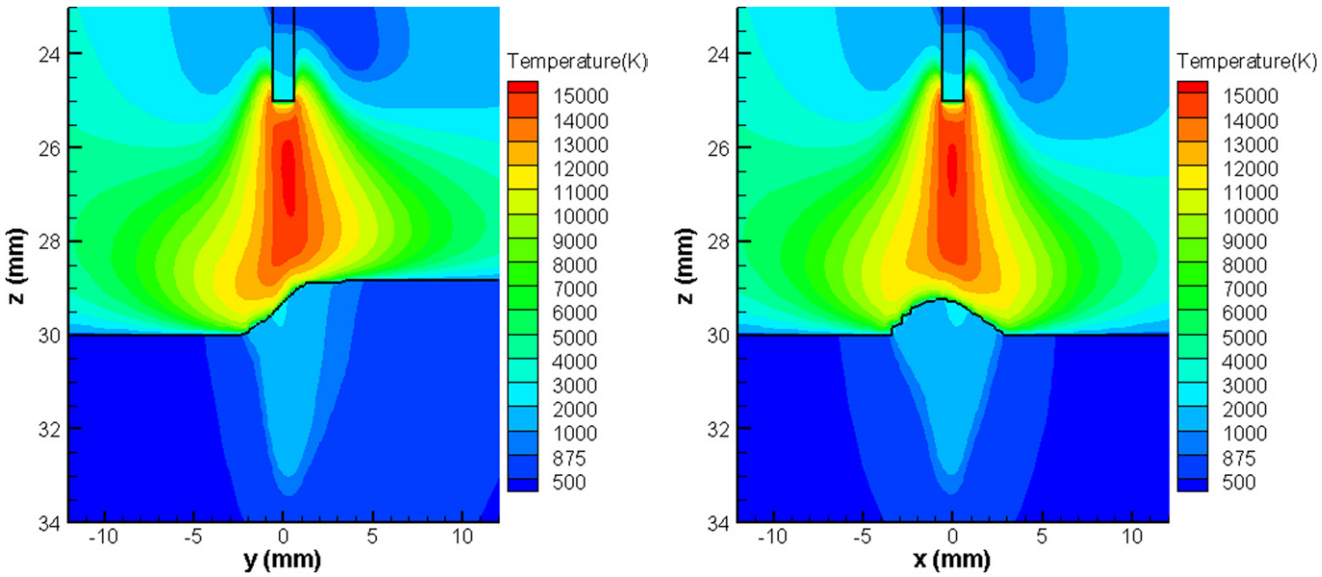


Figure 4. Temperature distribution in the cross-sections at $x = 0$ and $y = 0$. The welding velocity is to the left and into the page, respectively.

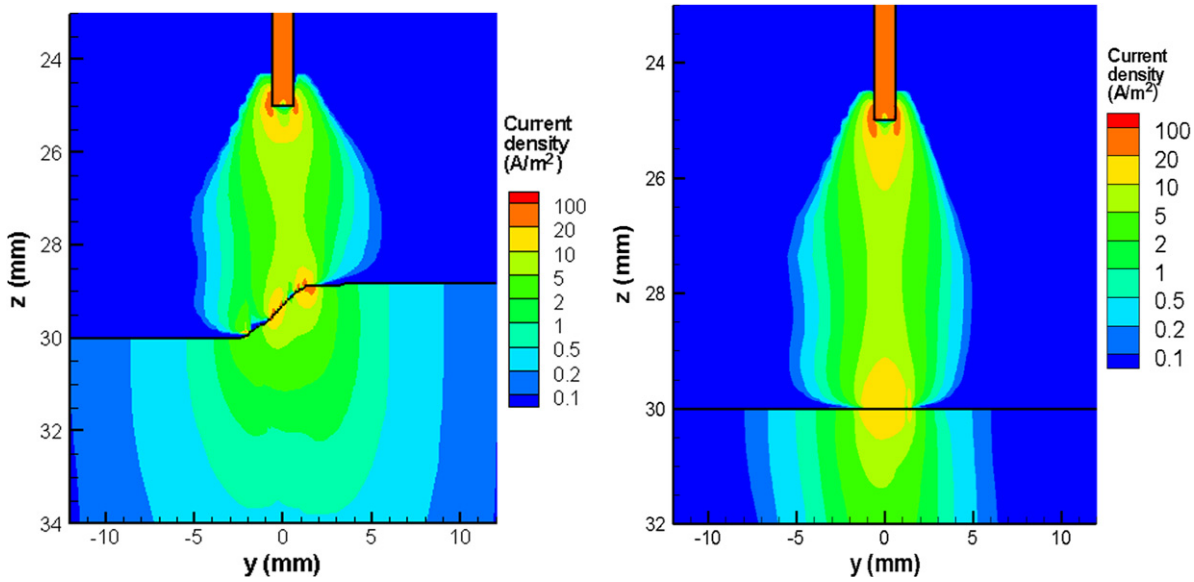


Figure 5. Current density distributions in the cross-section at $x = 0$ taking into account (left), and neglecting (right), the influence of weld-pool deformation. The welding velocity is to the left.

where I is the arc current, V_a is the arc voltage, r is the radial coordinate and T_w is the temperature at the top of the workpiece. Six arc parameters have to be determined empirically or otherwise estimated: the efficiency of energy transfer α, a convective heat transfer coefficient h_c, the arc force F, the arc temperature adjacent to the workpiece T_a, and effective radii for heat transfer and pressure, R_q and R_p, respectively.

It was noted in section 2 that the most important heating term at the weld-pool surface is the energy carried by electrons, which is proportional to the current density. This is illustrated by comparing figures 4 and 5; the region of the highest temperature in the weld pool is centred on the surface region for which the current density is greatest. Figure 5 indicates that, while a Gaussian distribution of current density and therefore heat flux may be reasonable for a flat workpiece, it is clearly grossly inaccurate for the real situation in which the weld pool is reinforced by the addition of droplets. This demonstrates the importance of including the arc in the computational domain.

3.2. Properties and influence of droplets

In addition to the transfer of metal that leads to the reinforcement of the weld pool, droplets also transfer energy and momentum to the weld pool. Figure 6 shows the change in the diameter, temperature and axial velocity of the droplets as they detach from the wire anode, fall through the arc, and enter the weld pool. Since evaporation of the droplets is neglected, their diameter remains constant in the arc. The droplets are accelerated by drag forces and by gravity, and heated convectively. When the droplets reach the weld pool, it is assumed they mix with the metal, spreading out as they approach the bottom of the weld pool. They transfer momentum and heat to the slower moving and cooler weld pool, again through the action of drag forces and convection.

Figure 7 shows the axial momentum and enthalpy source terms applied to the arc and weld pool to take into account the

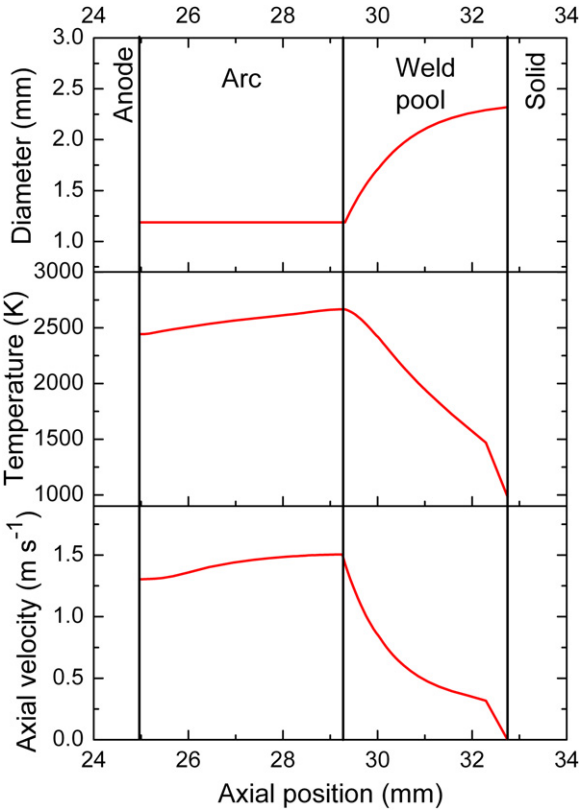


Figure 6. Diameter, temperature and axial velocity of a droplet as it travels through the arc and the weld pool. The height of the weld reinforcement is 0.7 mm at the position where the droplet reaches the weld pool, so the distance travelled through the arc is 4.3 mm.

influence of the droplets. The top of the graphs ($z = 25$ mm) corresponds to the position of the bottom of the wire, and the interface between the arc and weld pool is at $z = 29.2$ mm for $y \approx 0$ mm. As expected from figure 6, the source terms are negative in the arc (enthalpy and momentum are transferred from the arc to the droplets) and positive in the weld pool

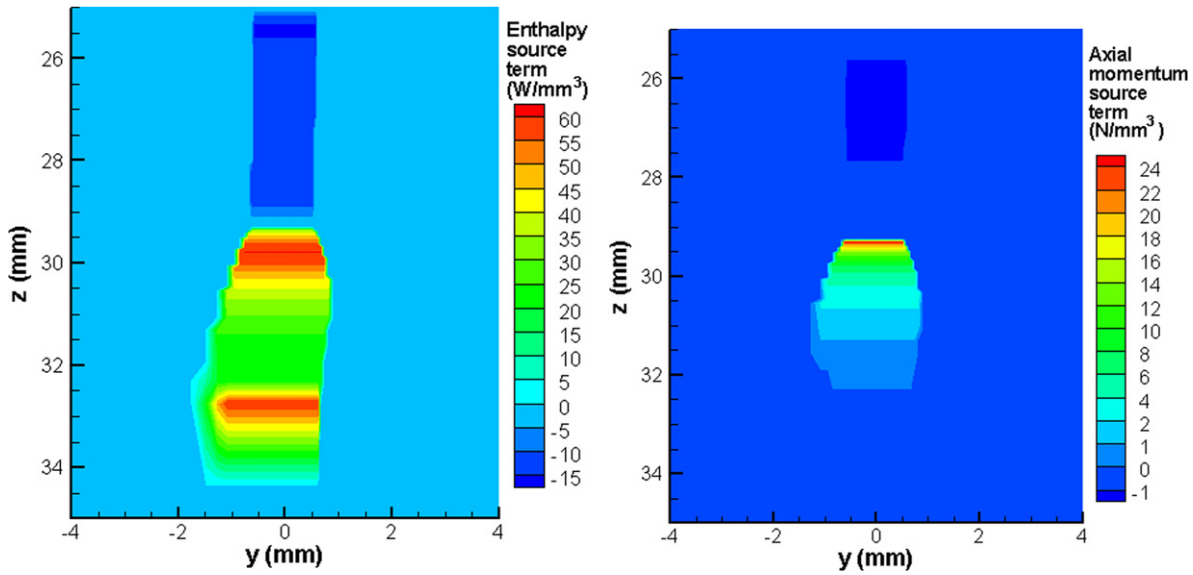


Figure 7. Enthalpy and axial momentum source terms applied to the arc and weld pool to take into account the influence of droplets. Results are shown in the $x = 0$ cross-section. The welding velocity is to the left.

(enthalpy and momentum are transferred from the droplets to the weld pool). The majority of the energy is transferred near the top and bottom of the weld pool, while the momentum transfer is more efficient, with most of the momentum being transferred near the top of the weld pool.

The flow in the weld pool is shown in figure 8. The direction of flow is determined by many factors, including the Marangoni effect (the change in surface tension with temperature), the magnetic pinch force, the viscous drag applied by the arc flow and buoyancy [2]. The position of the maximum in the momentum source term shown in figure 7 corresponds to the top of the region of strong downward flow in the weld pool. This indicates that the influence of droplet momentum is dominant near the top of the reinforced region, driving the liquid metal flow downwards and leading to a relatively deep weld pool. The magnetic pinch force also drives the flow downwards in this region, but its influence is less important. The Marangoni effect, which would leads to flow from regions of high temperature to low temperature at the weld-pool surface, has only a weak influence, as does the viscous drag applied by the arc flow and buoyancy.

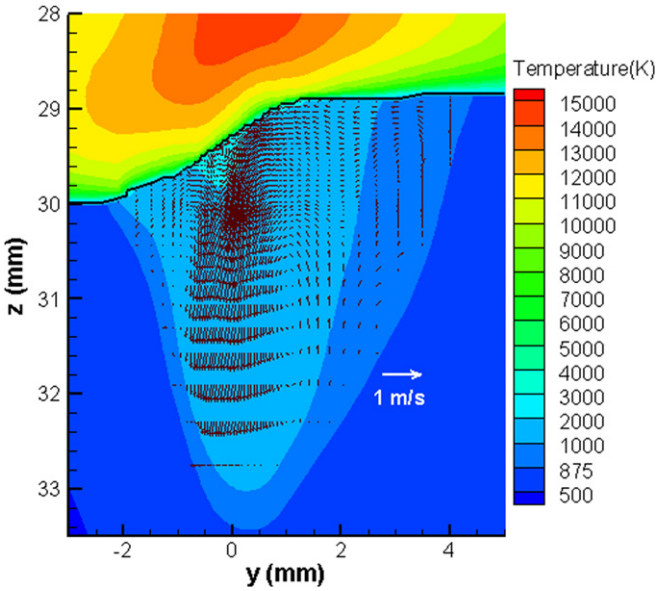


Figure 8. Temperature distribution and velocity vectors in the cross-section at $x = 0$, in the weld-pool region. The welding velocity is to the left. The white arrow near the right-hand edge of the weld pool shows the scale of the velocity vectors.

3.3. Comparison with measurements

A similar computational model to that presented here has been developed for gas–tungsten arc welding. The arc temperatures predicted by this model show good agreement with laser-scattering and spectroscopic measurements [2]. The presence of droplets makes such measurements much more difficult for GMAW, and as a consequence benchmarking has only been performed against measured weld cross-sections.

Figure 9 shows a measured weld cross-section for the same welding parameters that were used in the calculations. The cross-section was also calculated in the computational model, by determining the maximum extent of the molten region for all x–z planes. Note that the coarseness of the computational mesh leads to a straight line at the bottom of the calculated weld profile; in previous figures this has been smoothed by the plotting software.

The width of the weld and the height and shape of the reinforcement predicted by the model are in good agreement with measurements, but the predicted depth of the weld is substantially larger. The most likely reason for this discrepancy is that the presence of metal vapour is neglected in the calculation. Vaporization of the molten section of the wire and of the droplet will cool the droplet and reduce its volume, thereby decreasing the energy and momentum transferred by the droplet to the weld pool. Further, the presence of metal vapour will cool the arc. This will decrease the heat flux from the arc to the workpiece. All these changes will lead to a

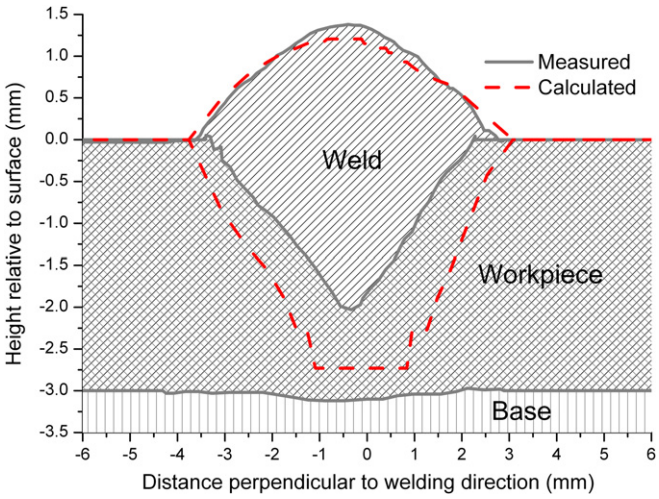


Figure 9. Comparison of measured and calculated weld profiles. The welding direction is into the page.

shallower weld pool. The influence of metal vapour on the arc properties is discussed in more detail in section 4.

4. Discussion

An important simplification in the model is that the time-dependent changes to the wire, arc and weld pool as droplets are formed at and detach from the wire, and then fall through the arc and into the weld pool, are not taken into account. While a completely comprehensive model would include such effects, this would highly increase the complexity and the running time of the computer code. The main goal of the work described here is to develop a model that can predict the shape and depth of the weld pool, so it is important to consider the influence of the simplification in this context.

The basis for the assumption that the influence of the droplets on the weld pool can be averaged in time is that the droplet frequency is large compared with the welding speed, and therefore the arc does not move far between successive droplets. For the parameters considered here, the arc moves 0.28 mm between droplets, much less than the droplet diameter of 1.19 mm, so the assumption is justified.

The changing shape of the wire electrode tip will affect the properties of the arc. Calculations have shown that there is an increase in the temperature of the arc near the electrode (by around 10%) as the wire tip changes shape from flat to hemispherical, due to the higher current density and consequent stronger ohmic heating. However, the weld-pool temperature is not significantly affected [1]. The axial flow velocity is much larger for the hemispherical tip. This is also a consequence of the higher current density near the tip, which increases the magnetic pinch force that drives convective flow. The higher flow velocity increases the arc pressure at the weld-pool surface [1]. While this will influence the surface profile of the weld pool, as shown by (12), for the arc current used here the flow velocities are sufficiently low for the arc pressure to have only a small effect. This was demonstrated in section 3.1, where it was shown that the weld-pool surface was everywhere convex. These considerations give confidence that the time-averaged approach used here will give acceptable accuracy.

The main phenomenon that has not been considered in the model is the influence of aluminium vapour, which is expected to be formed by evaporation of the molten region of the wire, and the droplets and weld pool. A comprehensive review of the influence of metal vapour in welding arcs has recently been published [34]. There is strong experimental [35–37] and theoretical [38–40] evidence that iron vapour has a dominant influence in GMAW arcs with steel electrodes, with a local temperature minimum occurring on the arc axis as a consequence of the strong radiation from iron atoms and ions. However, aluminium radiates much less strongly than iron. For example, the addition of 1% aluminium by volume to an argon plasma increases the net emission coefficient by about a factor of 10 for temperatures between 10 000 and 20 000 K, whereas the factor for iron is more than 1000 at 10 000 K, decreasing to around 10 at 20 000 K [41]. As reported by Lancaster [42], Smars and Acinger [43] found peak temperatures of up to 20 000 K occurring on axis in a 250 A GMAW arc with an aluminium wire electrode, confirming that the influence of aluminium vapour is less important than that of iron vapour.

Nevertheless, the presence of aluminium vapour will lead to increased radiative emission by the arc plasma, and therefore a decrease in the heat transferred to the workpiece. Further, vaporization of the electrode and droplets will cool the molten metal, so the energy transferred by the droplets to the weld pool will decrease. Thus the influence of metal vapour will lead to a shallower weld pool, in better agreement with measurements.

The results obtained have indicated the importance of including the arc plasma in the computational domain in a self-consistent manner. For example, the current density distribution at the weld-pool surface is strongly affected by the surface height, leading to a strong departure from the Gaussian distribution that is often assumed. Further, the droplets have a significant influence on the weld-pool flow, and their properties are influenced by the arc. Prediction of the influence of metal vapour will also require a model of the arc that allows the distribution of the vapour to be calculated, so as to determine the influence of the vapour on the temperature and current density distributions, and thus the heat flux to the weld pool [1, 44].

5. Conclusions

A three-dimensional computational model of GMAW that takes into account all important effects, except for the influence of metal vapour, has been developed. The model treats the wire electrode, arc and workpiece self-consistently, and allows the shape of the weld pool, including the surface deformation, to be predicted. The influence of the droplets on the arc and weld pool is taken into account on a time-averaged basis.

The model has allowed the prediction of temperatures, current densities and velocity distributions in the arc and weld pool, as well as the shape of the weld pool. The droplets are found to have an important influence on the weld pool. They of course increase the height of the weld-pool surface, which in turn influences the current density and heat flux distributions

here. Further, their energy and momentum lead to an increased weld-pool depth. These and other considerations show that accurate prediction of the weld-pool geometry requires that the arc plasma is included in the computational domain (rather than through boundary conditions at the weld-pool surface).

Reasonable agreement between the predicted and measured weld-pool cross-section is found. The main discrepancy is that the model overestimates the weld-pool depth; it is expected that including the effects of metal vapour will alleviate this problem.

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